Credit Risk Modeling

PD, LGD, EAD & Expected Loss — Basel IRB & Stress Testing

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Agenda

1. Project Overview
2. Portfolio EDA
3. Feature Selection
4. PD Modeling
5. Credit Scorecard
6. LGD Modeling
7. Expected Loss & Capital
8. Stress Testing
9. Fairness
10. Profitability
11. Conclusions

Banking Risk Management Pipeline

Why It Matters

Banks must accurately price credit risk to:

- Comply with **Basel III/IV** regulatory capital requirements
- Maintain solvency through stress periods
- Optimize **risk-adjusted returns** (RAROC)
- Demonstrate **fair lending** compliance

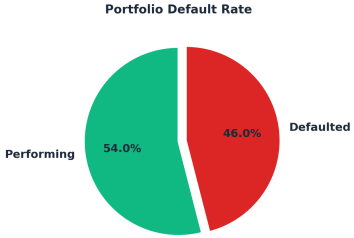
Approach

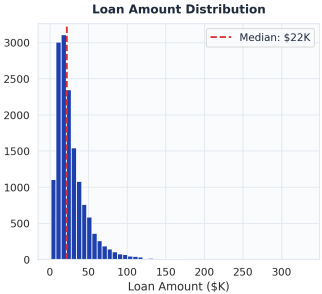
End-to-end credit risk pipeline: PD/LGD/EAD models → Expected Loss → RWA → Stress tests → Fairness audit.

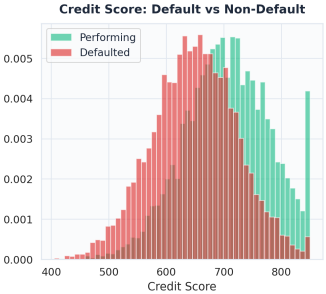
Portfolio

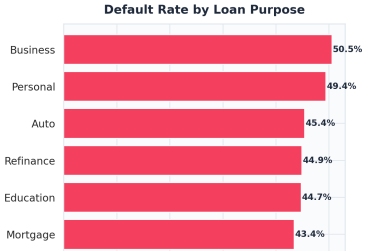
Loans	15,000
Total exposure	$375.4M
Default rate	38.0%
Expected Loss	$95.9M
Required capital	$67.8M

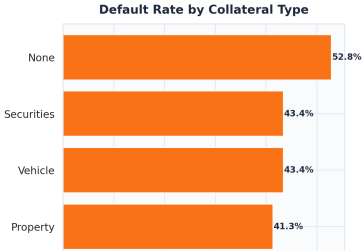
Portfolio Composition

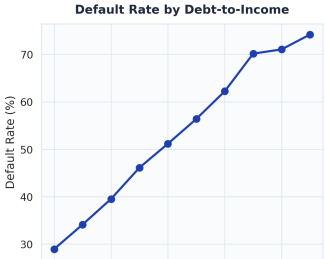




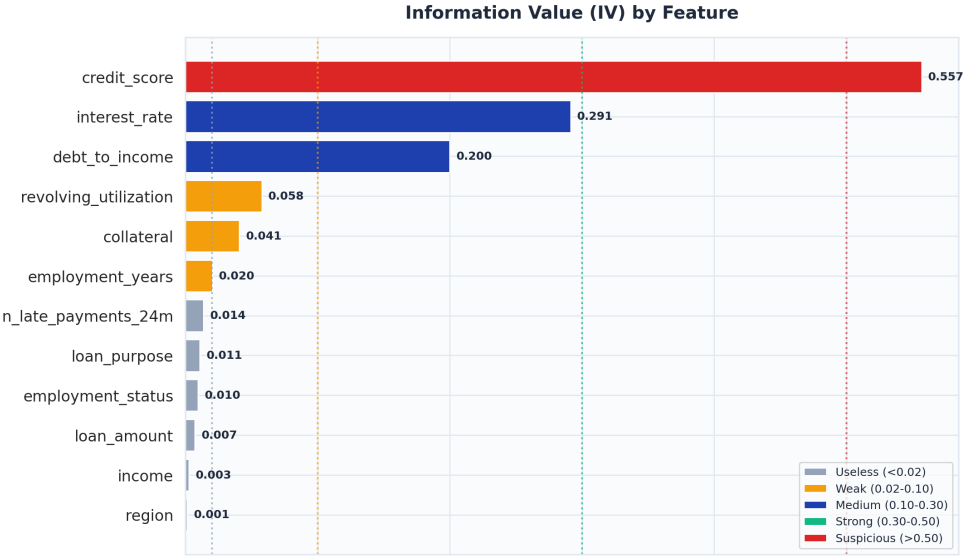




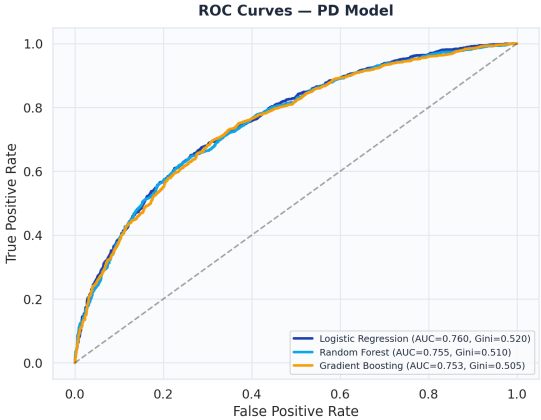


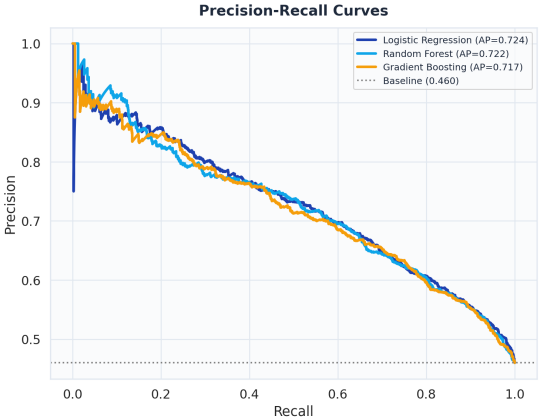


Information Value (IV) — Industry-Standard Feature Ranking



Probability of Default — Model Comparison

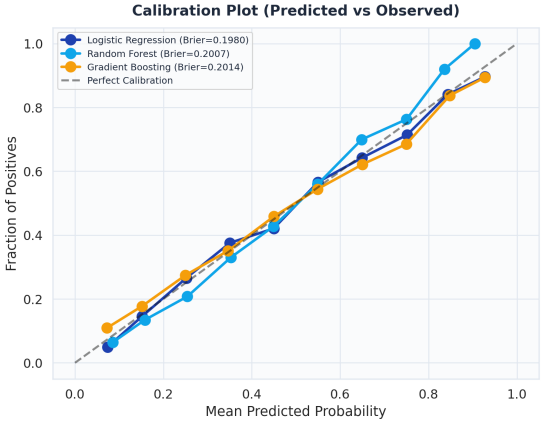


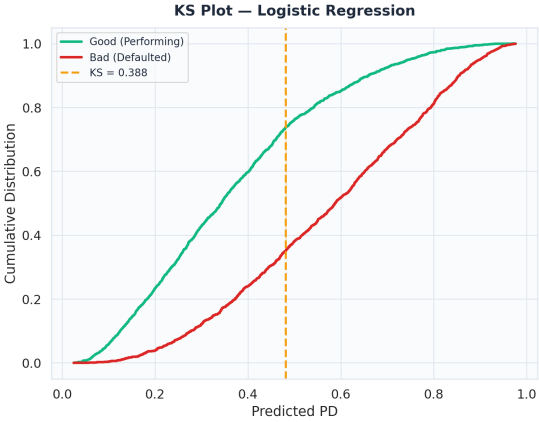


Industry Benchmark

Gini = 0.52, KS = 0.39 — meets the industry standard for production credit scorecards (Gini > 0.40, KS > 0.30). Logistic Regression remains preferred for **explainability** and

Calibration & KS Statistic

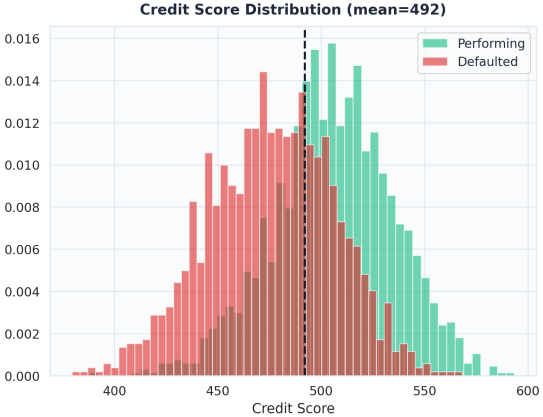


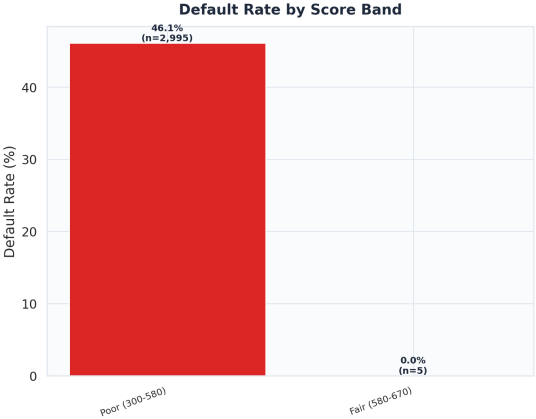


KS = max separation between Good and Bad cumulative distributions

Critical for setting cutoff thresholds in production scorecards.

Credit Scorecard — 300 to 850 Score Range

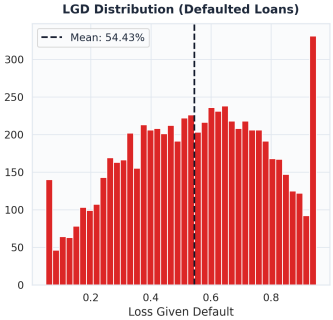


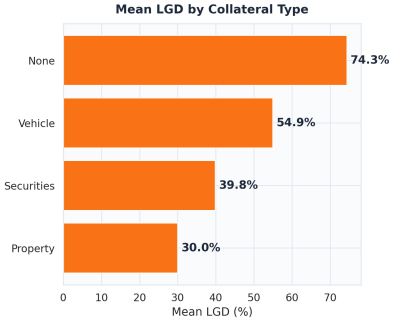


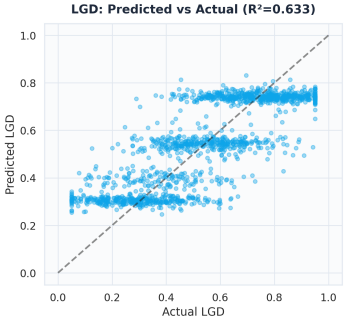
Standard Scorecard Mapping

Logistic Regression → score points using PDO=20 (Points to Double Odds), Base=600 at 50:1 odds. Default rates monotonically decrease across score bands.

Loss Given Default



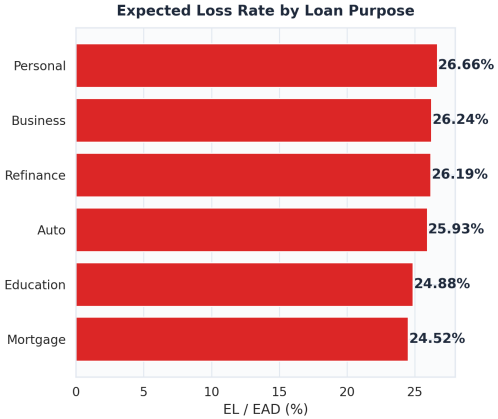


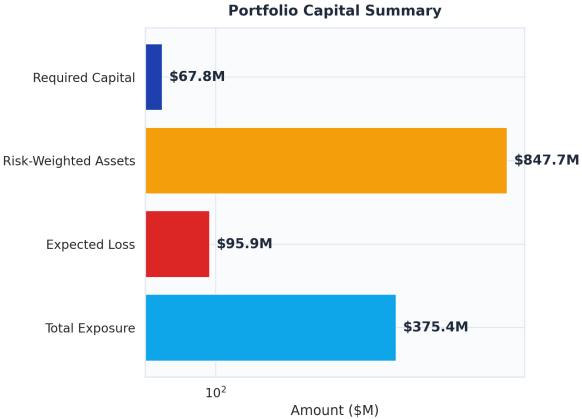


LGD Drivers

Property collateral reduces LGD to ∼30% vs **75% with no collateral**. Gradient Boosting achieves $R^2 = 0.63$ on held-out defaulted loans.

Expected Loss & Risk-Weighted Assets (Basel IRB)

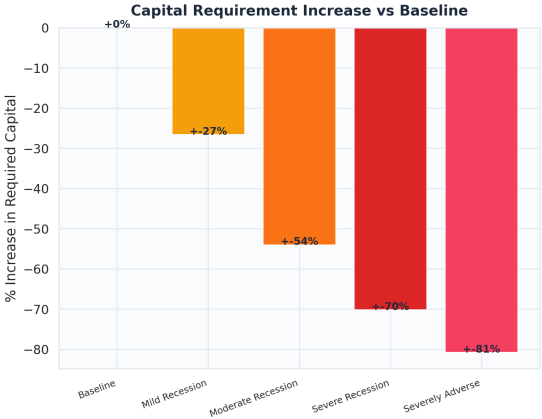




EL = PD × LGD × EAD

Personal and Business loans show the highest EL rates. Total Required Capital under Basel III: **$67.8M** (8% of RWA).

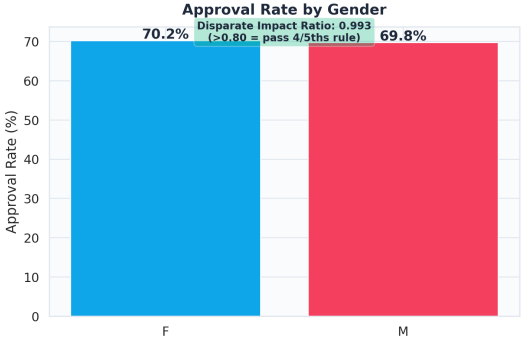
Stress Testing — 5 Regulatory Scenarios

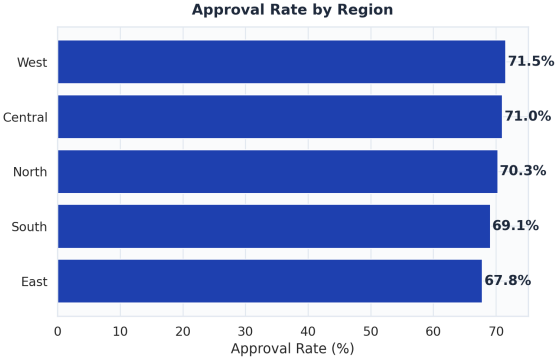


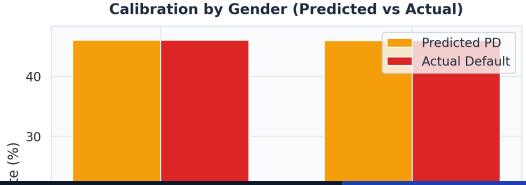
Severely Adverse Scenario

Under a 2008-style crisis (PD $\times 6$, LGD $+25$pp), Expected Loss grows from \$95.9M to \$270.1M. Bank must hold buffer capital to remain solvent.

Fair Lending Compliance



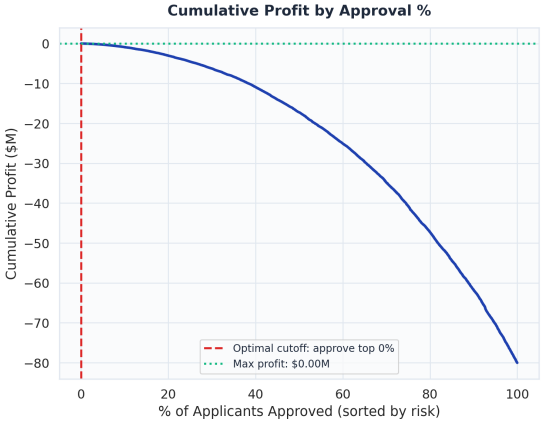


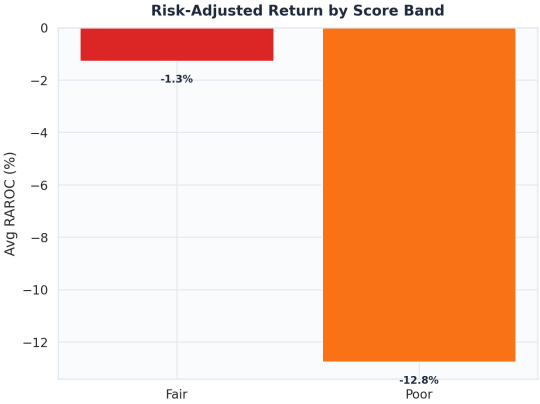


Fairness Metrics by Gender

Group	N	Approval Rate	Actual Default	Approved Default
F	7434	70.23	46.02	28.71

Profitability & Risk-Adjusted Return





Optimal Approval Cutoff

Cumulative profit peaks at a specific approval percentage — approving below this maximizes total profit. RAROC (Risk-Adjusted Return on Capital) varies dramatically by score band.

Executive Summary

Credit Risk Modeling — Executive Summary

PD Model Leaderboard

	AUC-ROC	Gini	KS	Brier	AP
Logistic Regression	0.7598	0.5196	0.3883	0.1980	0.7242
Random Forest	0.7551	0.5101	0.3840	0.2007	0.7223
Gradient Boosting	0.7527	0.5055	0.3853	0.2014	0.7173

```
PORTFOLIO SUMMARY

Loans:                    15,000
Total Exposure (EAD):     $375.4M
Default Rate:             46.00%
Mean LGD:                 52.6%

EXPECTED LOSS
Total EL:                 $95.90M
EL / EAD:                 25.55%

CAPITAL REQUIREMENTS
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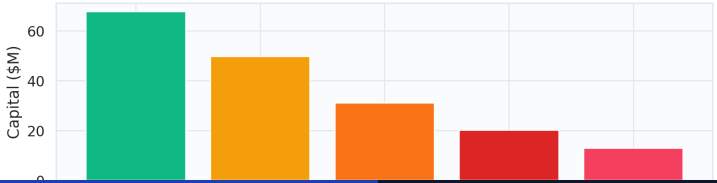
Required Capital Across Stress Scenarios



Production Roadmap & Future Work

Production Architecture

1. PD scorecard for **decisioning**
2. LGD/EAD models for **provisioning**
3. Quarterly **stress test** runs
4. Continuous **drift monitoring** (PSI)
5. Annual **fairness audits**
6. Model **governance & MRM**

Future Work

1. **Survival analysis** (time-to-default)
2. **Macroeconomic overlays** (TTC PD)
3. **SHAP explanations** (regulatory)
4. Counterfactual fairness audits
5. PIT vs TTC framework
6. Champion/Challenger in production

Thank You

Questions & Discussion

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